

Computer Science & IT

Discrete Mathematics



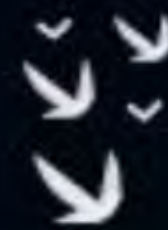
Combinatorics

Lecture No. 05



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Recap of Previous Lecture



Topic

Solution of recurrence relation using method of characteristic roots



Topic

Method of undetermined coefficient for particular solution



Topics to be Covered



Topic

Solution of recurrence relation using method of characteristic roots



Topic

Generating functions



H.W.



#Q. Find the solution of recurrence relation

$$a_n - a_{n-1} = n+2$$

$$\phi(E) = E - 1$$

$$\text{Char root} \Rightarrow t = 1$$

$$CF = C_1(1)^n$$

$$CF = C_1$$

$$f(n) = (n+2)$$

$$= (1)^n \cdot (n+2)$$

'1' is a root with multiplicity = '1'

∴ Particular Solution = $(1)^n (An+B)n^1$

$$P.S = n \cdot (An+B)$$

$$Q_n^{(P)} = P.S = n(An+B)$$

and it will satisfy the given recurrence relation

$$Q_n - Q_{n-1} = n+2$$

$$\therefore n \cdot (An+B) - (n-1)(A(n-1)+B) = n+2$$

$$\text{Put } n=0, \quad -(0-1)(A(-1)+B) = 2 \Rightarrow -A+B=2 \quad \left. \vphantom{\begin{array}{l} \text{Put } n=0, \\ -(0-1)(A(-1)+B) = 2 \end{array}} \right\} B=5/2$$

$$\text{Put } n=1, \quad (A+B) - 0 = \{ \quad \} = 1+2=3 \Rightarrow A+B=3 \quad \left. \vphantom{\begin{array}{l} \text{Put } n=1, \\ (A+B) - 0 = \{ \quad \} = 1+2=3 \end{array}} \right\} A=1/2$$

$$\therefore P.S = n \cdot \left(\frac{n}{2} + \frac{5}{2} \right)$$

Complete Soluⁿ $\Rightarrow Q_n = C.F + P.S.$

$$Q_n = C_1 + \frac{n^2}{2} + \frac{5n}{2}$$

#Q. Find the solution of recurrence relation

$$a_n - 2a_{n-1} + a_{n-2} = 3n+5$$

$$\phi(E) = E^2 - 2E + 1$$

$$\text{Ch. eqn} \Rightarrow t^2 - 2t + 1 = 0$$

$$\text{Ch. root} \Rightarrow \underbrace{t_1=1, t_2=1}_{\text{two equal roots}}$$

$$\begin{aligned} \text{C.F} &= (C_1 + C_2 n) \cdot (1)^n \\ &= C_1 + C_2 \cdot n \end{aligned}$$

$$f(n) = 3n + B$$

$$= (1)^n \cdot (3n + B)$$

'1' is a root with multiplicity=2

$$\therefore \text{P.S} = (1)^n \cdot (An + B) \cdot n^2$$

$$\boxed{\text{P.S} = n^2 \cdot (An + B)}$$

$$\underline{Q_n^{(P)} = n^2(A+B)} \quad Q_n - 2Q_{n-1} + Q_{n-2} = 3n + 5$$

$$n^2(A+B) - 2 \cdot (n-1)^2 \cdot (A(n-1)+B) + (n-2)^2 \cdot (A(n-2)+B) = 3n + 5$$

$$\text{We get, } A = 1/2$$

$$B = 4$$

$$\therefore PS = n^2 \cdot \left(\frac{n}{2} + 4 \right)$$

Complete Soluⁿ

$$Q_n = CF + PS \Rightarrow (C_1 + C_2 n) + \left(\frac{n^3}{2} + 4n^2 \right)$$

#Q. Find the solution of recurrence relation

$$a_n - a_{n-1} = 2^n \cdot n$$

Ch. root $\Rightarrow \boxed{t=1}$

$$f(n) = 2^n \cdot n$$

$$= 2^n (1 \cdot n + 0)$$

Polynomial of deg=1

'2' is root with multiplicity=0

$$\therefore P.S. = (2)^n \cdot (An+B) \cdot n^0$$

$$\boxed{P.S. = 2^n \cdot (An+B)}$$

$$Q_n^{(P)} = 2^n \cdot (An + B)$$

$$Q_n - Q_{n-1} = 2^n \cdot n$$

$$2^n \cdot (An + B) - 2^{(n-1)} \cdot (A(n-1) + B) = 2^n \cdot n$$

We get $A = 2$

$$B = -2$$

$$\therefore \text{P.S.} = 2^n (2n - 2)$$

Complete Soluⁿ

$$Q_n = \text{CF} + \text{PS} = \boxed{C_1 + 2^n (2n - 2)}$$

#Q. Find the solution of recurrence relation

$$a_n - 3.a_{n-1} = 3^n(n+2)$$

Ch. root $\Rightarrow t=3$

$$f(n) = 3^n(n+2)$$

'3' is a root with multiplicity = 1

$$\therefore P.S = (3)^n \cdot (An+B) \cdot n^1$$

$$P.S = n \cdot 3^n \cdot (An+B)$$

$$Q_n^{(p)} = n \cdot 3^n \cdot (An+B)$$

$$Q_n - 3Q_{n-1} = 3^n(n+2)$$

$$\therefore n \cdot 3^n(An+B) - 3 \left\{ (n-1) \cdot 3^{(n-1)} (A(n-1)+B) \right\} = 3^n(n+2)$$

$$A = 1/2$$

$$B = 5/2$$

$$\therefore PS = n \cdot 3^n \left(\frac{n}{2} + \frac{5}{2} \right) = 3^n \cdot \left(\frac{n^2}{2} + \frac{5n}{2} \right)$$

Complete Soluⁿ

$$Q_n = C_1 \cdot (3)^n + 3^n \left(\frac{n^2}{2} + \frac{5n}{2} \right)$$

Generating Function



Topic : Generating function

- Generating function is a way of generating an infinite sequence of numbers (a_n) by treating them as the coefficients of a formal power series.

$a_0, a_1, a_2, a_3, \dots$

Generating function
for given sequence
 $Gf(x)$

$$= a_0 \cdot x^0 + a_1 \cdot x^1 + a_2 \cdot x^2 + a_3 \cdot x^3 + \dots$$

Solution of this series is called
generating function in closed form

eg:- Consider the following sequence of real numbers.

1, 1, 1, 1, - - - - -

Find generating function for the above sequence in closed form.

Solu:-

Sequence = a_0 a_1 a_2 a_3 a_4
1, 1, 1, 1, 1, - - - - -

Generating function $Gf(x) = 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \dots$
(in the form of Power series) = $1 + x + x^2 + x^3 + \dots$

$$Gf(x) = \frac{1}{1-x}$$

first term
common ratio

it is an infinite GP with common ratio 'x'

Summation of infinite GP = $\frac{a}{1-r}$

$$Gf(x) = \frac{1}{(1-x)}$$

Consider the following sequence of real numbers.

$$a_0, a_1, a_2, a_3, \dots$$

Generating function

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \dots$$

$$Gf(x) = \sum_{i=0}^{\infty} (a_i) x^i$$

a_i is called general term

Binomial Expansion:-

$$\begin{aligned}(a+b)^n &= {}^nC_0 (a)^n (b)^0 + {}^nC_1 (a)^{n-1} (b)^1 + {}^nC_2 (a)^{n-2} (b)^2 + \dots + {}^nC_r (a)^{n-r} (b)^r + \dots + {}^nC_n (a)^0 (b)^n \\&= {}^nC_0 \cdot a^n \cdot b^0 + {}^nC_1 \cdot a^{n-1} \cdot b^1 + {}^nC_2 \cdot a^{n-2} \cdot b^2 + \dots + {}^nC_r (a)^{n-r} (b)^r + \dots + {}^nC_n (a)^0 (b)^n \\(1+x)^n &= {}^nC_0 (1)^n x^0 + {}^nC_1 (1)^{n-1} x^1 + {}^nC_2 (1)^{n-2} x^2 + \dots + {}^nC_r (1)^{n-r} x^r + \dots + {}^nC_n (1)^0 x^n\end{aligned}$$

$$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n$$

$$(1+x)^n = \sum_{i=0}^n ({}^nC_i) x^i$$

Q.12

Consider the following seq. of real numbers.

$$n_{c_0}, n_{c_1}, n_{c_2}, n_{c_3}, \dots, n_{c_n}$$

then what will be the generating function

Soln $Gf(x) = n_{c_0} \cdot x^0 + n_{c_1} \cdot x^1 + n_{c_2} \cdot x^2 + \dots + n_{c_n} \cdot x^n$

$$Gf(x) = (1+x)^n$$

eg:- Consider the following sequence of real numbers.
 $\overset{a_0}{1}, \overset{a_1}{2}, \overset{a_2}{3}, \overset{a_3}{4}, \overset{a_4}{5}, \dots$ } it is a sequence of natural nos.
Find the generating function for the above sequence

$$\text{Generating function } G_f(x) = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + 4 \cdot x^3 + \dots + (r+1) \cdot x^r + \dots$$

$$\frac{d}{dx}(x^n) = n \cdot x^{(n-1)}$$

$$\frac{1}{1-x} = 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \dots + 1 \cdot x^{r+1} + \dots$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx} \left\{ 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \dots + 1 \cdot x^{r+1} + \dots \right\}$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \left\{ 0 + 1 + 2 \cdot x + 3 \cdot x^2 + \dots + (r+1) \cdot x^r + \dots \right\}$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + \dots + (r+1) \cdot x^r + \dots$$

= Generating function for sequence 1, 2, 3, 4, - - -

eg:- Consider the following sequence of real numbers.

a_0 a_1 a_2 a_3 a_4
1, 2, 3, 4, 5, - - - - -

Find the generating function for the above sequence

Generating function
 $Gf(x) = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + 4 \cdot x^3 + \dots + (r+1) \cdot x^r + \dots$

$$Gf(x) = \frac{d}{dx} \left(\frac{1}{1-x} \right)$$

$$\frac{d}{dx} \left(\frac{p}{q} \right) = \frac{p'q - pq'}{q^2}$$

$$= \frac{\left(\frac{d}{dx}(1) \right) (1-x) - 1 \cdot \frac{d}{dx}(1-x)}{(1-x)^2} = \frac{0 - 1 \cdot (0-1)}{(1-x)^2}$$

$$Gf(x) = \frac{1}{(1-x)^2}$$

eg:- Find the Closed form of generating function.

$$Gf(x) = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \dots$$

Sequence = 0, 1, 2, 3, ... { it is a sequence of whole number }

Soln.

$$Gf(x) = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \dots + r \cdot (x)^r + \dots$$

We know, $\frac{1}{(1-x)^2} = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + \dots + (r+1) \cdot x^r + \dots$

multiply L.H.S & R.H.S by x

$$\frac{x}{(1-x)^2} = 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \dots + (r+1) \cdot x^{r+1} + \dots$$

if we multiply it by 'x' then it becomes $(r+1) \cdot x^{r+1}$

$$\frac{x}{(1-x)^2} = 0 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 3 \cdot x^3 + \dots + (r+1) \cdot x^{r+1} + \dots$$

∴ Generating function w.r.t. 0, 1, 2, 3, 4, ... is $\frac{x}{(1-x)^2}$

* If we want to right shift a sequence by 'K' position then multiply the sequence by x^K .

∴ The generating function for sequence

0, 0, 0, 1, 1, 1, 1, 1, 1, 1, - - - - - will be

Right shift
by '3' position
∴ multiply by x^3

generating function is
 $\left(\frac{1}{1-x}\right)$

Hence generating fn for given seq. will be $x^3 \cdot \left(\frac{1}{1-x}\right) = \left(\frac{x^3}{1-x}\right)$

Binomial Coefficient

We know, $nC_r = \frac{n!}{r!(n-r)!} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-r+1) \cdot \overbrace{(n-r)(n-r-1)\dots 3 \cdot 2 \cdot 1}^{(n-r)!}}{(r \cdot (r-1)(r-2)\dots 3 \cdot 2 \cdot 1) \cdot \cancel{(n-r)!}}$

$$nC_r = \frac{(n-0) \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-(r-1))}{r!}$$

from 0 to $(r-1)$ we have
' r ' terms { we perform subtraction }
from 0 to $(r-1)$

Binomial Coefficient

from '0' to $(r-1)=3$

$${}^6C_4 =$$

$$\frac{(6-0) \cdot (6-1) \cdot (6-2) \cdot (6-3)}{4!}$$

$r=4,$
 $\therefore (r-1) = (4-1) = 3$

$$= \frac{\overset{3}{\cancel{6}} * 5 * \cancel{4} * \cancel{3}}{\cancel{4} * \cancel{3} * \cancel{2} * 1} = \textcircled{15} \underline{\underline{\text{Ans}}}$$

Extended Binomial Coefficient

Find the value of $^{-5}C_3$

$$^{-5}C_3 = \frac{(-5-0)(-5-1)(-5-2)}{3!} = \frac{-5 * \cancel{-6} * -7}{\cancel{6}}$$

$$r = 3$$

$$(r-1) = (3-1) = 2$$

∴ Subtract from '0' to '2'

$$= (-5) \cdot (-1) \cdot (-7)$$

$$= -35$$

Ans

Extended Binomial Coefficient

We know $nCr = \frac{n!}{r!(n-r)!}$

Find the value of $-nCr$ $\left\{ \begin{array}{l} \text{where 'n' \& 'r' are} \\ \text{non-negative integers} \end{array} \right\}$

$$-nCr = \frac{(-n-0) \cdot (-n-1) \cdot (-n-2) \cdot \dots \cdot (-n-(r-1))}{r!}$$

$$\text{Subtract from } 0 \text{ to } (r-1) = (-1)^r \cdot \underline{(n+0) \cdot (n+1) \cdot (n+2) \cdot \dots \cdot (n+(r-1))}$$

$$= (-1)^r \cdot \frac{(n+r-1) \cdot (n+r-2) \cdot \dots \cdot (n+2) \cdot (n+1) \cdot n}{r!}$$

$$\frac{(n-1) \cdot (n-2) \cdot \dots \cdot 3 \cdot 2 \cdot 1}{(n-1) \cdot (n-2) \cdot \dots \cdot 3 \cdot 2 \cdot 1}$$

$$= (-1)^r \cdot \frac{(n+r-1)!}{r! (n-1)!} =$$

$$(-1)^r \frac{(n+r-1)!}{r! (n-r)!} = (-1)^r \binom{n+r-1}{r}$$

$$-n_{G_r} = (-1)^r (n+r-1)_{G_r}$$

$$-n C_r = (-1)^r (n+r-1) C_r$$



Topic : Extended binomial coefficient

$$-nC_r = (-1)^r \cdot (n+r-1)C_r$$

Where 'n' & 'r' are non-negative integers.

Q:- Find the Coefficient of x^r w.r.t.
following generating functions.

$$\textcircled{1} \quad Gf(x) = (1+x)^n = nC_0 \cdot x^0 + nC_1 \cdot x^1 + nC_2 \cdot x^2 + \dots + nC_r \cdot x^r + \dots$$

$$Gf(x) = \sum_{i=0}^{\infty} \binom{n}{i} x^i$$

Coefficient of $x^r = nC_r$ it is the general term it gives the coefficient of x^i

$$\textcircled{2} \quad Gf(x) = (1-x)^{-n}$$

$$= (1+(-x))^{-n} = -nC_0 \cdot (-x)^0 + -nC_1 \cdot (-x)^1 + -nC_2 \cdot (-x)^2 + \dots + -nC_r \cdot (-x)^r + \dots$$

$$Gf(x) = \sum_{i=0}^{\infty} \binom{n+i-1}{i} x^i$$

Coefficient of $x^r = (n+r-1)C_r$ general term coefficient of x^i

$$\begin{aligned} -nC_r \cdot (-x)^r &= \{-nC_r\} \{(-1)^r \cdot x^r\} \\ &= \{(-1)^r \cdot n+r-1\} \cdot \{(-1)^r \cdot x^r\} \\ &= (-1)^{2r} (n+r-1) C_r \cdot x^r \\ &= 1 \cdot (n+r-1) C_r \cdot x^r \end{aligned}$$

H.W. ③ $Gf(x) = (1-x)^n =$

Coefficient of $x^r = ?$

H.W. ④ $Gf(x) = (1+x)^{-n}$

Coefficient of $x^r = ?$

H.W.

$$\textcircled{5} \quad Gf(x) = (1+ax)^n =$$

Coefficient of $x^r = ?$

H.W.

$$\textcircled{6} \quad Gf(x) = (1-ax)^{-n}$$

Coefficient of $x^r = ?$

H.W

⑦ $Gf(x) = (1 - ax)^n =$

H.W Coefficient of $x^r = ?$

H.W ⑧ $Gf(x) = (1 + ax)^{-n}$

Coefficient of $x^r = ?$



2 mins Summary



✓ **Topic**

Solution of recurrence relation using method of characteristic roots

✓ **Topic**

Generating functions

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THANK - YOU